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Lessons from a Sentinel-2 project in Ifrane National Park, Morocco

Working with an AI agent in QGIS

Scoping it, keeping it moving and checking what it produces, for any GIS or remote sensing project.

Author
Abdelaali MOUID

Course
Spatial Industry X.0, Pr Badr BEN HICHOU

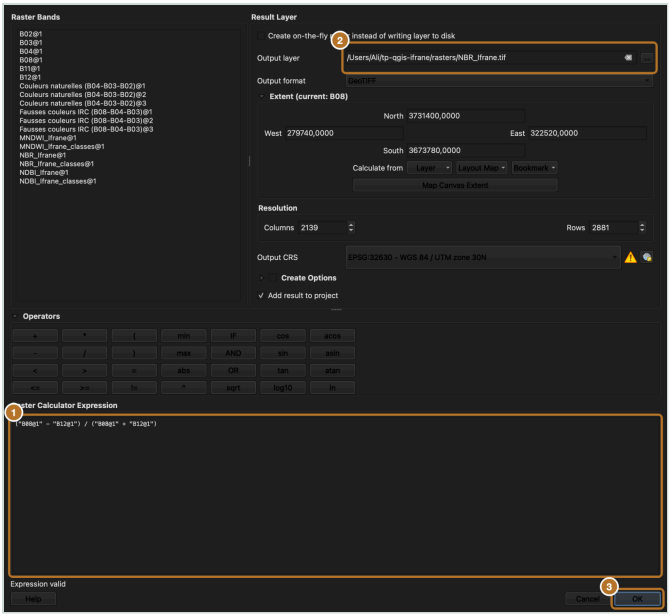
Why this guide

For a remote sensing project in QGIS (Sentinel-2 indices, classes, areas and maps over the Ifrane forest in Morocco), I worked with Claude running as an agent on my computer. The hands-on work, which would have taken me days by hand, was done quickly, and every result could be checked. This guide collects the way of working I recommend after the fact. It does not walk through an exercise: the method applies to any QGIS task.

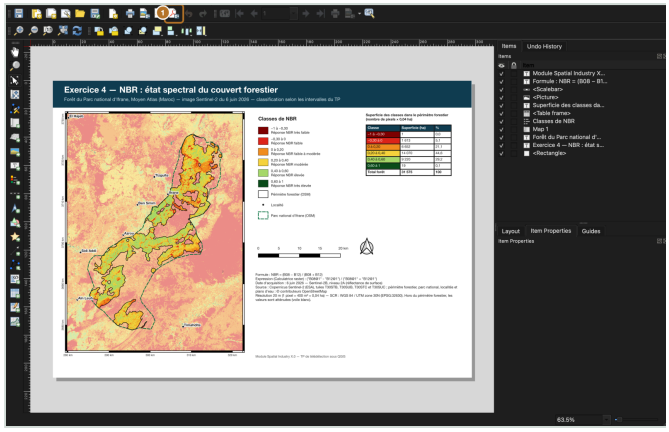
What the agent really does

It does not click around in QGIS. It writes small Python programs that use the QGIS engine (the PyQGIS interface) and its processing tools, runs them, looks at what comes out, then fixes things if needed. In practice these are the same tools as in the menus: raster calculator, reprojection, zonal statistics, print layouts. The agent can also open QGIS to show you a filled-in dialog, a map or a project.

Your role does not disappear, it changes. You ask the question, set the rules, judge the results and decide what goes into the final document. The agent does the hands-on work, often better and faster than by hand, but it does not know what makes sense on the ground.



The QGIS raster calculator, filled in by the agent, then opened for checking.



A generated print layout, as it appears in the project.

> **In one sentence**
>
> Used well, the agent makes the hands-on work almost free. The effort moves to scoping at the start and checking at the end, and those are the two moments this guide covers in detail.

Setting up

One folder per project

Open the agent on a dedicated folder holding everything that matters: the assignment, your data, your area boundaries. Keep the raw data separate and never let the agent modify it, so a failed step can restart from it.

```
my-project/
├── sources/     assignment, papers
├── data/raw/    original data
├── data/        prepared data
├── scripts/     numbered steps
├── outputs/     maps and tables
└── CLAUDE.md    project rules
```

The project rules file

The agent automatically reads a rules file placed at the root of the folder (`CLAUDE.md` for Claude Code, `AGENTS.md` for Codex), at the start of every session. This is the trick that saved me the most time: you write down once what you would otherwise repeat in every request.

```
# Project rules
- QGIS 3.44 installed in /Applications/QGIS.app. Use its Python (PyQGIS).
- Area: forest of Ifrane National Park. CRS: EPSG:32630 (UTM 30N).
- Sentinel-2 level 2A, bands converted to reflectance, 20 m grid.
- Keep each step in a numbered script in scripts/. Never modify data/raw/.
- QGIS projects with relative paths.
- Before handing me a result: check value ranges, projection
  and resolution, and show me a preview of the maps.
- Flag any deviation from the assignment.
```

The whole method also exists as a skill the agent opens by itself whenever a task involves QGIS, with templates, checking scripts and tested recipes: `github.com/abdelaalimouid/qgis-agent`.

Let the agent find the data

There is no need to download everything by hand. The agent can query open catalogues and read only the area it needs, which is much faster than downloading whole tiles.

Data	Where the agent gets it
Sentinel-2, Landsat	Open STAC catalogues (Earth Search on AWS, Microsoft Planetary Computer), no account needed
Elevation	Copernicus DEM (30 m)
Boundaries, roads, water	OpenStreetMap, through the Overpass API
Rainfall, climate	CHIRPS, ERA5 (for time series)

> **OpenStreetMap is not an official source**
>
> The park outline there covers about 60,000 ha, against the 125,000 ha usually quoted. Whenever an area figure matters, ask where each boundary comes from.

Writing the request

The agent takes your words literally. A good request reads like a brief for a capable colleague who has not taken your course.

Item	What to specify	Example
Goal	What you want to find out or produce	Condition of the cedar forest and open water
Area	The place and where its boundaries come from	Ifrane National Park, OSM outline
Data	Sensor, period, cloud constraints	Sentinel-2 L2A, May to June 2026, 0 % clouds
Required method	Formulas, thresholds and tools the assignment requires	The assignment's formulas and intervals, raster calculator
Deliverables	Expected files and their format	Rasters, PDF maps, area table, .qgz project
Checks	What must be verified before handing in	Value ranges, previews, deviations from the assignment

Request template

```
Read the assignment in sources/. Goal: [question]. Area: [place, origin of the boundaries].
Data: [sensor, period, clouds]. Follow the assignment's formulas and intervals exactly.
Deliverables: [list]. Before starting, propose a short plan and wait for my approval.
At the end, check the results, show me a preview of the maps and list deviations from the
assignment.
```

Three habits that change everything

- Attach the assignment or instructions rather than summarising them. The agent finds details in it you would have forgotten, such as the unit for areas or the required map elements.
- Also say what it must not do: change the area, alter the thresholds, overwrite a file.
- Ask for a plan before anything runs. Fixing a plan takes a minute; fixing a result can take thirty.

Short follow-ups, once the frame is set

```
Rerun the same chain on the 11 June image and compare the class areas with 6 June.
Dayet Aaoua comes out at 0 ha of water: show it to me in natural colour before I write that.
List everything that departs from the assignment and everything you assumed without checking.
Explain in three sentences why NDBI is the exact opposite of NDMI here.
The legend on map 5 overflows: put the labels on two lines and change nothing else.
```

Each one does a single thing: rerun the chain, check a figure against the image, bring out the assumptions, understand a result, fix a detail without touching the rest. Ask the third one on every project before handing in: it is the one that surfaces the deviations and limits you need to document.

The work cycle that works

Five stages, always in the same order. They apply to a single reprojection as much as to a full processing chain.

1

Get a plan approved

The agent lists the steps, the QGIS tools it intends to use, the files it will create and the open questions. You correct it before a single line runs. This is the moment to check the projection, the resolution and the area the calculations cover.

2

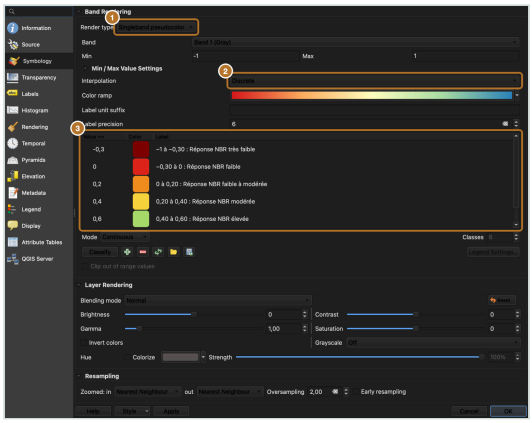
Run it end to end

Once the plan is approved, let the agent work through it without stopping it at every step. It can run a download in the background and write the next step while it waits. Ask it to keep each step in a numbered script: that is what lets you rerun everything later.

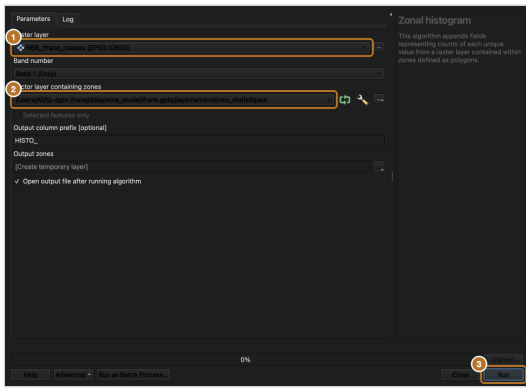
3

Ask for evidence

Don't settle for "done". Ask for a preview of each map, the key figures, a screenshot of the dialog that was used. The agent can open QGIS, fill in the tool with its parameters and take a screenshot: you see exactly what was done.



Discrete symbology with the assignment's classes.



Zonal histogram used to count pixels per class.

4

Correct in small steps

One precise remark at a time: "the legend overflows", "redo the map with the scale in kilometres", "exclude the ocean from the ring". Short corrections take a few minutes. If something breaks, paste the exact error message or a screenshot: the agent finds the cause much faster than from a description.

5

Get a work report

At the end, ask for a factual account of what was done and what came out of it. It is the base for everything written next, by you or by the agent. The next part details what it should contain and how to get the write-up.

What the agent does very well in QGIS

Repetitive, long or technical tasks are where the gain is clearest.

Task	Example request
Find an image	Find the least cloudy Sentinel-2 image over my area between April and June 2026 and check clouds and snow over the area itself.
Prepare rasters	Reproject to EPSG:32630, clip to the area, bring every band to 20 m.
Compute indices	Compute NBR with the raster calculator and the exact formula from the assignment.
Classify and measure	Classify with these intervals and give the area and percentage of each class in the forest.
Map series	Make one print layout per index with title, legend, scale bar, north arrow, date and source, exported to PDF.
Time series	Repeat the same calculation on every clear date since 2016 and plot the change for each lake.
Supervised classification	With my training points, train a random forest and give the confusion matrix.
Combine with other layers	For each low-value patch, give its area and the nearest town.
Make the work clickable	Turn the chain into a QGIS graphical model so it can be rerun without code.
Debug and explain	Here is the QGIS error: find the cause. Explain what this parameter does.

> **My case**
>
> Switching study areas (from the Maâmora forest to Ifrane) took one sentence: the agent reused the scripts, found a cloud-free date across the four tiles involved, changed the projection and remade the maps. That is the point of keeping every step as a script.

Checking without redoing everything

The agent rarely gets calculations wrong; assumptions are wrong more often. These checks take little time and catch most problems.

Check	How
Redo one step by hand	Recompute an index and its classes in QGIS and compare. On my first project, in the Maâmora forest, the NDVI redone by hand was identical pixel for pixel (11,353,377 pixels).
Look at the image	Display the natural colour composite under every surprising result. A "dry" lake should look dry.
Orders of magnitude	A normalised index stays between −1 and 1; class areas must add up to the area total.
Reopen the project	Open the .qgz file: what is saved can differ from what was exported. I found legends there that listed every layer.
Projection and resolution	Check the CRS and pixel size of every layer: a wrong UTM zone shifts everything.
Deviations and assumptions	Ask for the list of what departs from the assignment and what was assumed without being checked.

Warning signs

- A uniform or nearly empty map, or a class at 0 % that should exist.
- Values outside the expected range, or areas that do not add up.
- A figure that changes between the table and the text.
- A relationship that is too weak or too strong between two linked quantities. A regression slope of 0.13 instead of 0.92 came from a calculation in single precision; the gap with the class means gave it away.

> **Classic remote sensing mistakes**
>
> Indices computed on raw values instead of reflectance; 10 m and 20 m bands mixed; tiles taken on different dates; wrong UTM zone; in discrete symbology, the lower bound entered instead of the upper bound; areas computed over the whole rectangle instead of the study area. Write them into your rules file so the agent checks them by itself.

The work report, then the write-up

Ask the agent for a work report first: a factual document, like a lab notebook. It says what was done, with what, and what came out. Any report, paper or summary is then written from it, by you or by the agent.

Starting from facts keeps the write-up honest: every figure in the final text can be traced back to a table, and the limits stay visible instead of being smoothed over.

What it should contain

Section	Content
Purpose	The question addressed and the area
Data	Sensor, date, tiles, bands, resolution, boundary sources, licences
Processing	Each operation with its QGIS tool and the parameters used
Outputs	The list of files produced
Results	Figures and tables, without comment
Checks	What was verified and how
Limits and open questions	What remains uncertain and what deserves a closer look

> The request
>
> ```
> Give me a factual work report: data used, processing steps with their parameters, files
> produced, results in tables, checks done, limits and open questions.
> ```

A real example from my project is in the appendix.

Asking for the write-up

> The request
>
> ```
> From the work report, write the full report: introduction, method, results, discussion,
> conclusion. Use only figures from the results, keep the limits visible, and mark what I need
> to check with my own knowledge of the area.
> ```

Then read it as its author: check each figure against the tables and rewrite what you cannot defend.

In the paper

Declare the help you received in a few lines, in the methods or the acknowledgements depending on the journal: the tool used, the tasks it was given and how the results were checked. An AI cannot be an author, and you remain responsible for every figure. Credit the data too: "Contains modified Copernicus Sentinel data (2026)", "© OpenStreetMap contributors, ODbL licence".

Making the work reusable

- Keep the numbered scripts: with them, rerunning on another date or another area takes one sentence.
- Save QGIS projects with relative paths, so the folder can be moved or zipped.
- Record the versions (QGIS, GDAL) in the rules file.
- Keep the list of your requests: it is the record of your method choices.
- For a paper, publish scripts and results with an identifier (Zenodo, for example), and offer a QGIS model for readers who don't code.

Checklist

Before handing in an assignment or submitting a paper.

Before starting

- [ ] Dedicated folder, raw data protected
- [ ] Rules file up to date
- [ ] Assignment or instructions attached
- [ ] Plan approved: tools, projection, resolution, area

During

- [ ] Every step kept in a script
- [ ] A preview of every map requested and looked at
- [ ] Corrections one at a time

Before handing in

- [ ] One step redone by hand and compared
- [ ] Surprising results checked on the image
- [ ] QGIS project reopened
- [ ] Deviations from the assignment listed and justified
- [ ] Work report collected, write-up checked against it
- [ ] Help declared, data credited

Installing the qgis-agent skill

It is published at `github.com/abdelaalimouid/qgis-agent`. In Claude Code, two commands are enough:

```
/plugin marketplace add abdelaalimouid/qgis-agent
/plugin install qgis-agent@qgis-agent
```

Otherwise, download `qgis-agent.zip` from the repository's Releases and unzip it into `~/.claude/skills/`, or `~/.agents/skills/` for Codex. Then start a new session on the project folder. QGIS 3 or 4 must be installed.

> First run
>
> `Use the qgis-agent skill: check that QGIS is detected, then propose a project rules file for this project.`

Appendix: a real work report

From the Ifrane project, exercises 4 to 6 (NBR, MNDWI, NDBI). It contains only facts and figures; the analysis is written from it.

Work report, Ifrane forest, NBR, MNDWI, NDBI — November 2026

Data

Image	Sentinel-2B level 2A, 6 June 2026, tiles T30STB, T30SUB, T30STC, T30SUC (same pass). Clouds, shadows and snow over the area: below 0.01 %.
Bands	B02, B03, B04, B08 (averaged from 10 to 20 m), B11, B12. Reflectance = value × 0.0001 − 0.1, clamped to [0, 1].
Area	OSM forest polygons inside the OSM park outline: 31,575 ha. Ring from 0 to 3 km: 64,608 ha. Processing area: park + 5 km. EPSG:32630, 20 m pixel (0.04 ha).

Processing

Raster calculator (the assignment's formulas); discrete pseudocolour symbology, the assignment's six classes; reclassification into classes 1 to 6; zonal histogram over the forest and the ring; three print layouts exported to PDF; NDVI and NDMI computed as a complement; cross-check with OpenStreetMap water bodies, towns and built-up areas.

Results

Indicator	Forest	Ring
Mean NBR	0.294	0.080
NBR < 0	5.1 % (1,613 ha)	37.0 %
NBR from 0.20 to 1	73.8 %	19.7 %
MNDWI > 0 (open water)	0.8 ha	179.8 ha
Mean NDBI	−0.098	0.043
NDBI > 0	20.2 % (6,365 ha)	73.6 %
Mean NDVI	0.544	0.310

Water body (OSM)	Mapped	Wet on 6 June
Lake Afnourir	87.8 ha	72.1 ha (82 %)
Dayet Maskar	52.4 ha	39.5 ha (75 %)
Ben Smim dam	10.7 ha	8.6 ha (81 %)
Sidi Mimoun dam	6.9 ha	1.7 ha (25 %)
Lake Ifrah	118.6 ha	7.2 ha (6 %)
Dayet Aaoua	102.8 ha	0 ha
Dayet Hachlaf	28.7 ha	0 ha

Correlations in the forest: NBR and NDVI r = 0.93; NBR and NDMI r = 0.99; NDBI and NDMI r = −1.00. Mean NDBI within 1 km: Ifrane −0.14, Azrou 0.00, El Hajeb 0.05.

Checks

- Clouds, shadows and snow checked on the scene classification layer (SCL) over the area.
- Pixel counts per class equal the number of pixels in each zone (789,381 in the forest, 1,615,200 in the ring).
- OpenStreetMap water bodies: 31 % of their pixels have a positive MNDWI.
- Negative NBR patches, Lake Afnourir, Dayet Aaoua, Ifrane and Azrou inspected in natural colour.
- QGIS project reopened: layers, styles and three print layouts present.

Limits and open questions

- A single date, no field data; approximate OpenStreetMap outlines.
- Since when have Dayet Aaoua and Dayet Hachlaf been dry, and does the link with rainfall hold?
- Are the negative NBR patches in the north-east (20 to 90 ha) grassland, logging or dieback?
- Are the plots with NDBI above 0.20 outside the forest (8 to 20 ha) ploughed soil or quarries?